

Physics · Medium paper

Real VCAA questions ranked by the published state results. Editable: type your answers straight into this PDF, save it, then mark yourself with the guide at the back.

Name _____ Date _____

122 marks

reading 15 min · writing 150 min

State average on these questions

72 / 122 (59%)

a student at the state average on every question

How to use this paper

1. Open each question with its blue link. It opens the official VCAA exam PDF at the right page. The question text isn't reproduced here because it is VCAA copyright (some exams include third-party material).
2. Answer in this file: click a bubble for multiple choice, and type in the boxes for written answers (long answers scroll inside the box). Methods and science working can be typed in plain text, e.g. $v = \sqrt{2gh} = 9.9 \text{ m/s}$.
3. Keep to the real time limits above, and don't look at the guide until you've finished.
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5. Mark yourself: in the guide at the back, tick each marking point you earned, enter your marks, and compare your total with the state average.

Why these questions

Every question is a real VCAA exam question, chosen by ranking all the questions from the listed exams by how Victorian students actually scored on them (percentage correct for multiple choice, state average ÷ marks for written questions). This paper takes the middle band of questions.

Sources

- [2022 Physics exam \(PDF\)](#) · [examiners' report](#) (*previous study design*)
- [2023 Physics exam \(PDF\)](#) · [examiners' report](#) (*previous study design*)
- [2024 Physics exam \(PDF\)](#) · [examiners' report](#)
- [2025 Physics exam \(PDF\)](#) · [examiners' report](#)

Section A – Multiple-choice questions

Open each question in the official exam PDF (click the blue link), then select A–D. One mark each.

Question 1 · 1 mark

2022 VCAA exam · Section A · Question 5 · [previous study design](#) · [open in the official exam PDF \(p. 5\)](#) ↗

Output when an alternator's rotation frequency doubles

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 2 · 1 mark

2022 VCAA exam · Section A · Question 8 · [previous study design](#) · [open in the official exam PDF \(p. 7\)](#) ↗

Compression needed to store 0.9 J in a spring

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 3 · 1 mark

2022 VCAA exam · Section A · Question 13 · [previous study design](#) · [open in the official exam PDF \(p. 9\)](#) ↗

Wave speed from a standing wave on a rope

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 4 · 1 mark

2022 VCAA exam · Section A · Question 14 · [previous study design](#) · [open in the official exam PDF \(p. 9\)](#) ↗

Evidence for electrons behaving as waves

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 5 · 1 mark

2023 VCAA exam · Section A · Question 1 · [previous study design](#) · [open in the official exam PDF \(p. 2\)](#) ↗

Magnetic poles in a loudspeaker from the force direction

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 6 · 1 mark

2023 VCAA exam · Section A · Question 5 · [previous study design](#) · [open in the official exam PDF \(p. 5\)](#) ↗

When a globe lights as a magnetic field is switched off

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 7 · 1 mark

2023 VCAA exam · Section A · Question 6 · [previous study design](#) · [open in the official exam PDF \(p. 5\)](#) ↗

Magnetic flux through a coil

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 8 · 1 mark

2023 VCAA exam · Section A · Question 8 · [previous study design](#) · [open in the official exam PDF \(p. 6\)](#) ↗

Two people of different mass falling together

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 9 · 1 mark

2023 VCAA exam · Section A · Question 10 · [previous study design](#) · [open in the official exam PDF \(p. 8\)](#) ↗

Spring constant from a force–compression graph

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 10 · 1 mark

2023 VCAA exam · Section A · Question 19 · [previous study design](#) · [open in the official exam PDF \(p. 11\)](#) ↗

Frequency of a visible hydrogen spectral line

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 11 · 1 mark

2024 VCAA exam · Section A · Question 4 · [open in the official exam PDF \(p. 4\)](#) ↗

Launch speed from a spring's force–compression graph

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 12 · 1 mark

2024 VCAA exam · Section A · Question 5 · [open in the official exam PDF \(p. 5\)](#) ↗

Total energy of a falling rock over time

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 13 · 1 mark

2024 VCAA exam · Section A · Question 18 · [open in the official exam PDF \(p. 14\)](#) ↗

Fringe spacing when L doubles and d halves

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 14 · 1 mark

2024 VCAA exam · Section A · Question 19 · [open in the official exam PDF \(p. 15\)](#) ↗

Linear graph for orbital speed and radius

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 15 · 1 mark

2025 VCAA exam · Section A · Question 1 · [open in the official exam PDF \(p. 2\)](#) ↗

Tow-rope tension on an accelerating water skier

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 16 · 1 mark

2025 VCAA exam · Section A · Question 5 · [open in the official exam PDF \(p. 6\)](#) ↗

Field directions around a point mass and a negative charge

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 17 · 1 mark

2025 VCAA exam · Section A · Question 10 · [open in the official exam PDF \(p. 9\)](#) ↗

Generator current for an ideal step-up transformer

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 18 · 1 mark

2025 VCAA exam · Section A · Question 12 · [open in the official exam PDF \(p. 11\)](#) ↗

Choosing the trace for 240 V RMS at 50 Hz

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 19 · 1 mark

2025 VCAA exam · Section A · Question 14 · [open in the official exam PDF \(p. 12\)](#) ↗

Which traveller is not in an inertial frame

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 20 · 1 mark

2025 VCAA exam · Section A · Question 19 · [open in the official exam PDF \(p. 15\)](#) ↗

Why an acceleration–force graph has a non-zero intercept

Your answer: ☐ A ☐ B ☐ C ☐ D

Section B – Short-answer questions

Open each question in the official exam PDF, then type your answer to every part in the boxes. Show working where more than one mark is available.

Question 1 (9 marks)

2022 VCAA exam · Section B · Question 2 · [previous study design](#) · [open in the official exam PDF \(p. 14\)](#) ↗

Geostationary satellites: why over the equator, altitude, speed

a. Why geostationary satellites must be above the equator (2 marks)

b. Show the altitude is 3.59×10^7 m (4 marks)

c. Orbital speed of a geostationary satellite (3 marks)

Question 2 (7 marks)

2023 VCAA exam · Section B · Question 1 · [previous study design](#) · [open in the official exam PDF \(p. 12\)](#) ↗

Two charged balls hanging from strings: forces and tension

a. Draw T, Fg and FE on each ball (3 marks)

b. Show the tension is $4.5 \times 10^{-2} \text{ N}$ (2 marks)

c. Magnitude of the electrostatic force (2 marks)

Question 3 (7 marks)

2023 VCAA exam · Section B · Question 2 · [previous study design](#) · [open in the official exam PDF \(p. 14\)](#) ↗

Phobos orbiting Mars: field strength, period, decaying orbit

a. Show g at 6000 km altitude is 0.48 N kg^{-1} (2 marks)

b. Orbital period of Phobos in seconds (3 marks)

c. Does the period get shorter or longer as Phobos moves closer? (2 marks)

Question 4 (3 marks)

2023 VCAA exam · Section B · Question 6 · [previous study design](#) · [open in the official exam PDF \(p. 20\)](#) ↗

DC generator with the magnets on the axis of rotation

a. Why this magnet orientation generates no EMF (2 marks)

b. Where to place the magnets to generate an EMF (1 mark)

Question 5 (3 marks)

2023 VCAA exam · Section B · Question 11 · [previous study design](#) · [open in the official exam PDF \(p. 30\)](#) ↗

Guitar string standing wave

a. Frequency of the wave shown (fundamental) (1 mark)

b. How the standing wave forms on the fixed string (2 marks)

Question 6 (6 marks)

2023 VCAA exam · Section B · Question 13 · [previous study design](#) · [open in the official exam PDF \(p. 32\)](#) ↗

Young's double slit: path difference, red laser, wave evidence

a. Path difference $S_1P_2 - S_2P_2$ in metres (2 marks)

b. Effect of a red laser on the band spacing (1 mark)

c. How the experiment supports a wave model, not a particle model (3 marks)

Question 7 (6 marks)

2023 VCAA exam · Section B · Question 14 · [previous study design](#) · [open in the official exam PDF \(p. 34\)](#) ↗

Neutron beam: speed from wavelength, diffraction, electron comparison

a. Speed of the neutrons (2 marks)

b. Will a crystal of spacing 3.62×10^{-10} m diffract the beam? (2 marks)

c. Electron or neutron with the same wavelength: which is faster? (2 marks)

Question 8 (8 marks)

2023 VCAA exam · Section B · Question 15 · [previous study design](#) · [open in the official exam PDF \(p. 36\)](#) ↗

Calcium photoelectric graph, then copper

a. Work function of calcium from the graph (1 mark)

b. Maximum wavelength that ejects electrons (2 marks)

c. Draw the line for copper (work function 4.70 eV) (2 marks)

d. Does 380 nm light eject electrons from copper? (3 marks)

Question 9 (6 marks)

2023 VCAA exam · Section B · Question 16 · [previous study design](#) · [open in the official exam PDF \(p. 38\)](#) ↗

Mercury spectral line and energy levels

a. Photon energy for the 436.6 nm line (2 marks)

b. Draw the transition for the 436.6 nm line (1 mark)

c. All photon energies emitted after absorbing 6.7 eV (3 marks)

Question 10 (7 marks)

2024 VCAA exam · Section B · Question 3 · [open in the official exam PDF \(p. 20\)](#) ↗

Golf ball hit off a cliff: height, range, air resistance

a. Height of the cliff (landing 6.2 s after being hit) (3 marks)

b. Horizontal distance travelled (2 marks)

c. How air resistance would change the flight path (2 marks)

Question 11 (4 marks)

2024 VCAA exam · Section B · Question 4 · [open in the official exam PDF \(p. 22\)](#) ↗

Crash-test dummy: impulse from a force–time graph, airbag

a. Impulse on the dummy in the initial test (2 marks)

b. Sketch the force–time graph with the airbag (double the duration) (2 marks)

Question 12 (3 marks)

2024 VCAA exam · Section B · Question 6 · [open in the official exam PDF \(p. 25\)](#) ↗

Could the ISS’s gravity hold a supply craft in orbit?

Evaluate the suggestion with calculations (3 marks)

Question 13 (6 marks)

2024 VCAA exam · Section B · Question 7 · [open in the official exam PDF \(p. 26\)](#) ↗

Electron accelerated then curved in a magnetic field; proton comparison

a. Show the electron's speed is $1.32 \times 10^7 \text{ m s}^{-1}$ (1 mark)

b. Accelerating voltage needed (2 marks)

c. Path of a proton at the same speed, justified (3 marks)

Question 14 (8 marks)

2024 VCAA exam · Section B · Question 15 · [open in the official exam PDF \(p. 41\)](#) ↗

Electron microscope: momentum, diffraction prediction, X-ray comparison

a. Momentum of an electron in the beam (2 marks)

b. Who is right about a detectable diffraction pattern? (2 marks)

c. How electrons and X-rays can give almost identical patterns (2 marks)

d. Energy of the X-ray photons in eV (2 marks)

Question 15 (6 marks)

2025 VCAA exam · Section B · Question 8 · [open in the official exam PDF \(p. 28\)](#) ↗

Observation satellite: period, gravitational force, why it accelerates

a. Show the period is 6.62×10^3 s (2 marks)

b. Gravitational force on the satellite (2 marks)

c. Why the satellite accelerates towards Earth (Newton's laws) (2 marks)

Question 16 (7 marks)

2025 VCAA exam · Section B · Question 12 · [open in the official exam PDF \(p. 34\)](#) ↗

Hand-cranked generator: flux change, average EMF, DC output

a. Why the flux through the coil changes as it rotates (1 mark)

b. Show the flux change from horizontal to vertical is 7.5×10^{-4} Wb (1 mark)

c. Average EMF over the quarter turn at 50 Hz (3 marks)

d. Change to produce DC output, with reason (2 marks)

Question 17 (3 marks)

2025 VCAA exam · Section B · Question 16 · [open in the official exam PDF \(p. 40\)](#) ↗

Ek max against frequency graph: Planck's constant and work function

a. Planck's constant from the graph (2 marks)

b. Work function from the graph (1 mark)

Question 18 (3 marks)

2025 VCAA exam · Section B · Question 17 · [open in the official exam PDF \(p. 41\)](#) ↗

Rocket vehicle: speed from two measured time intervals

Rachel's speed relative to Sandra, as a multiple of c (3 marks)

END OF PAPER

Marking guide follows

Stop here until you have finished the paper under exam conditions.

Marking guide

Section A – Multiple-choice questions

✓	Q	Ans	Why	How Victoria went
☐	1	C	Double the frequency: the period halves and the peak EMF doubles (flux changes twice as fast). <i>Most popular wrong answer: B (33%).</i>	62% correct A 1% · B 33% · C 62% · D 4%
☐	2	B	Stored energy = area under the force–compression graph. Find the compression where the triangle's area reaches 0.9 J (about 0.06 m).	72% correct A 14% · B 72% · C 9% · D 5%
☐	3	A	Read the wavelength from the loops over 2.4 m ($\lambda = 1.2$ m), then $v = \lambda/T = 1.2/1.6 = 0.75$ m s ⁻¹ . <i>Most popular wrong answer: C (27%).</i>	63% correct A 63% · B 4% · C 27% · D 4%
☐	4	D	Electron diffraction through a crystal is wave behaviour. The photoelectric effect and spectra don't show electron waves.	68% correct A 12% · B 12% · C 7% · D 68%
☐	5	A	Apply the right-hand rule with the current direction and the force into the page to find the field direction, and hence the poles. <i>Most popular wrong answer: B (25%).</i>	65% correct A 65% · B 25% · C 7% · D 3%
☐	6	A	Current is induced only while the flux is changing: off before, on during the switch-off, off after. <i>Most popular wrong answer: B (24%).</i>	59% correct A 59% · B 24% · C 9% · D 8%
☐	7	A	$\Phi = BA = 0.2 \times \pi(0.05)^2 \approx 1.6 \times 10^{-3}$ Wb. Flux does not depend on the number of loops. <i>Most popular wrong answer: B (22%).</i>	59% correct A 59% · B 22% · C 15% · D 3%
☐	8	D	Different weights, but $a = F/m = g$ for both, so they land together because the acceleration is the same. <i>Most popular wrong answer: B (19%).</i>	68% correct A 6% · B 19% · C 8% · D 68%
☐	9	D	Gradient of the line: $k = \frac{40 \times 10^3 \text{ N}}{8.0 \times 10^{-3} \text{ m}} = 5.0 \times 10^6$ N m ⁻¹ . Watch the kN and mm.	71% correct A 8% · B 14% · C 6% · D 71%
☐	10	B	Read $\lambda \approx 650$ nm from the spectrum: $f = c/\lambda = 3.0 \times 10^8 / 6.5 \times 10^{-7} \approx 4.6 \times 10^{14}$ Hz. <i>Most popular wrong answer: C (19%).</i>	70% correct A 5% · B 70% · C 19% · D 5%
☐	11	C	Spring PE = area = $\frac{1}{2}(0.10)(40) = 2.0$ J = $\frac{1}{2}mv^2$, so $v = \sqrt{2 \times 2.0/0.050} = 8.9$ m s ⁻¹ . <i>Most popular wrong answer: B (17%).</i>	63% correct A 13% · B 17% · C 63% · D 8%
☐	12	A	With no air resistance, mechanical energy is conserved: the total is a horizontal line.	73% correct A 73% · B 7% · C 10% · D 10%
☐	13	D	$\Delta x = \lambda L/d$: doubling L and halving d multiplies the spacing by 4.	63% correct A 13% · B 12% · C 11% · D 63%
☐	14	D	$v = \sqrt{GM/r}$, so $v^2 = GM \cdot \frac{1}{r}$: plotting v^2 against $1/r$ gives a straight line through the origin.	69% correct A 3% · B 13% · C 14% · D 69%
☐	15	D	$T - 500 = 65 \times 4.0$, so $T = 760$ N. <i>Most popular wrong answer: A (17%).</i>	60% correct A 17% · B 15% · C 8% · D 60%

✓	Q	Ans	Why	How Victoria went
☐	16	A	Gravitational fields always point towards the mass, and the field around a negative charge points inwards too. <i>Most popular wrong answer: D (17%).</i>	68% correct A 68% · B 14% · C 2% · D 17%
☐	17	D	Power in = power out: $I = P/V = 100 \times 10^6 / 20 \times 10^3 = 5.0 \times 10^3$ A. <i>Most popular wrong answer: C (16%).</i>	70% correct A 5% · B 8% · C 16% · D 70%
☐	18	D	Peak = $240\sqrt{2} \approx 340$ V and period = $1/50 = 0.02$ s. Pick the trace with both. <i>Most popular wrong answer: B (25%).</i>	65% correct A 6% · B 25% · C 3% · D 65%
☐	19	D	Going around a bend at constant speed is centripetal acceleration, so that frame is non-inertial. <i>Most popular wrong answer: A (18%).</i>	60% correct A 18% · B 5% · C 16% · D 60%
☐	20	C	Acceleration with zero applied force means an extra constant force acts. A bench sloping down in the direction of motion adds a gravity component.	66% correct A 14% · B 9% · C 66% · D 11%

Multiple-choice score: _____ / 20

Section B – Short-answer questions

Question 1 (9 marks) · state average 4.8 / 9 (53%)

2022 VCAA exam · Section B · Question 2

a. Why geostationary satellites must be above the equator

- ☐ The orbit must be centred on Earth's centre (gravity points to the centre)
- ☐ To stay above one point it must orbit in the same plane and about the same axis as Earth's rotation, which is the equatorial plane

Answer / notes: Centred on Earth and in the plane of Earth's spin. Only 2% got full marks.

State average 0.4 / 2 (20%); 2% scored full marks. Mark distribution: 0: 67% · 1: 31% · 2: 2%

Your mark: _____ / 2

b. Show the altitude is 3.59×10^7 m

- ☐ $\frac{GMm}{r^2} = \frac{4\pi^2 mr}{T^2}$
- ☐ $r = \sqrt[3]{\frac{GMT^2}{4\pi^2}}$
- ☐ $r = 4.22 \times 10^7$ m
- ☐ Altitude = $r - R_E = 3.59 \times 10^7$ m

Answer / notes: Subtract Earth's radius at the end.

State average 2.5 / 4 (62%); 56% scored full marks. Mark distribution: 0: 29% · 1: 6% · 2: 4% · 3: 4% · 4: 56%

Your mark: _____ / 4

c. Orbital speed of a geostationary satellite

- ☐ $v = \frac{2\pi r}{T}$ with $r = 4.22 \times 10^7$ m
- ☐ Correct substitution
- ☐ $v \approx 3.07 \times 10^3$ m s⁻¹

Answer / notes: $2\pi(4.22 \times 10^7)/86\,400 = 3.07 \times 10^3$ m s⁻¹. Using Earth's radius was the common error.

State average 1.9 / 3 (63%); 54% scored full marks. Mark distribution: 0: 19% · 1: 24% · 2: 3% · 3: 54%

Your mark: _____ / 3

Question 2 (7 marks) · state average 4.3 / 7 (61%)

2023 VCAA exam · Section B · Question 1

a. Draw T, Fg and FE on each ball

- ☐ Tension along each string towards the ceiling
- ☐ Weight vertically down
- ☐ Electric force horizontal, away from the other ball (repulsion)

Answer / notes: Three arrows on each ball, starting at the ball.

State average 2.4 / 3 (80%); 56% scored full marks. Mark distribution: 0: 6% · 1: 8% · 2: 29% · 3: 56%

Your mark: _____ / 3

b. Show the tension is 4.5×10^{-2} N

- ☐ Each string is 30° from the vertical: $T \cos 30^\circ = mg$
- ☐ $T = \frac{0.0040 \times 9.8}{\cos 30^\circ} = 4.5 \times 10^{-2}$ N

Answer / notes: The vertical component of T balances the weight.

State average 1.1 / 2 (55%); 53% scored full marks. Mark distribution: 0: 41% · 1: 5% · 2: 53%

Your mark: _____ / 2

c. Magnitude of the electrostatic force

- ☐ $F_E = T \sin 30^\circ$ (or $mg \tan 30^\circ$)
- ☐ $F_E \approx 2.3 \times 10^{-2}$ N

Answer / notes: $0.045 \times 0.5 = 0.023$ N

State average 0.8 / 2 (40%); 39% scored full marks. Mark distribution: 0: 58% · 1: 3% · 2: 39%

Your mark: _____ / 2

Question 3 (7 marks) · state average 4.1 / 7 (59%)

2023 VCAA exam · Section B · Question 2

a. Show g at 6000 km altitude is 0.48 N kg^{-1}

- ☐ $g \propto 1/r^2$ with $r = 3390 + 6000 = 9390$ km
- ☐ $3.72 \times (3390/9390)^2 = 0.48 \text{ N kg}^{-1}$

Answer / notes: Use a ratio with the radius from the centre, not the altitude.

State average 1.2 / 2 (60%); 58% scored full marks. Mark distribution: 0: 39% · 1: 3% · 2: 58%

Your mark: _____ / 2

b. Orbital period of Phobos in seconds

- ☐ $g = \frac{4\pi^2 r}{T^2}$, so $T = 2\pi\sqrt{r/g}$
- ☐ $r = 9.39 \times 10^6$ m and $g = 0.48$ substituted
- ☐ $T \approx 2.8 \times 10^4$ s

Answer / notes: $2\pi\sqrt{9.39 \times 10^6 / 0.48} = 2.8 \times 10^4$ s

State average 1.6 / 3 (53%); 40% scored full marks. Mark distribution: 0: 26% · 1: 30% · 2: 4% · 3: 40%

Your mark: _____ / 3

c. Does the period get shorter or longer as Phobos moves closer?

- ☐ Shorter
- ☐ $T = 2\pi\sqrt{r^3/GM}$: a smaller orbital radius means a smaller period

Answer / notes: Shorter.

State average 1.3 / 2 (65%); 49% scored full marks. Mark distribution: 0: 21% · 1: 29% · 2: 49%

Your mark: _____ / 2

Question 4 (3 marks) · state average 1.8 / 3 (60%)

2023 VCAA exam · Section B · Question 6

a. Why this magnet orientation generates no EMF

- ☐ The field is parallel to the plane of the loop at all times, so there is no flux through it
- ☐ As the loop rotates the flux doesn't change, so by Faraday's law no EMF is induced

Answer / notes: No flux, so no change in flux, so no EMF.

State average 0.9 / 2 (45%); 32% scored full marks. Mark distribution: 0: 41% · 1: 27% · 2: 32%

Your mark: _____ / 2

b. Where to place the magnets to generate an EMF

- ☐ Rotated 90° so the field is perpendicular to the axis of rotation (poles on either side of the loop)

Answer / notes: Across the loop, not along the axle.

State average 0.9 / 1 (90%); 88% scored full marks. Mark distribution: 0: 12% · 1: 88%

Your mark: _____ / 1

Question 5 (3 marks) · state average 1.7 / 3 (57%)

2023 VCAA exam · Section B · Question 11

a. Frequency of the wave shown (fundamental)

- ☐ $\lambda = 2L = 1.5 \text{ m}$, so $f = v/\lambda = 262 \text{ Hz}$

Answer / notes: 262 Hz

State average 0.7 / 1 (70%); 67% scored full marks. Mark distribution: 0: 33% · 1: 67%

Your mark: _____ / 1

b. How the standing wave forms on the fixed string

- ☐ Travelling waves reflect from the fixed ends
- ☐ The incident and reflected waves superpose (interfere) to give fixed nodes and antinodes (when L is a whole number of half-wavelengths)

Answer / notes: Reflection plus superposition.

State average 1.0 / 2 (50%); 32% scored full marks. Mark distribution: 0: 39% · 1: 29% · 2: 32%

Your mark: _____ / 2

Question 6 (6 marks) · state average 3.5 / 6 (58%)

2023 VCAA exam · Section B · Question 13

a. Path difference $S_1P_2 - S_2P_2$ in metres

- ☐ Second bright band from the centre: path difference = 2λ
- ☐ $2 \times 510 \times 10^{-9} = 1.02 \times 10^{-6} \text{ m}$

Answer / notes: $1.02 \times 10^{-6} \text{ m}$. "1020 m" (unit slip) should have looked absurd.

State average 0.9 / 2 (45%); 41% scored full marks. Mark distribution: 0: 54% · 1: 5% · 2: 41%

Your mark: _____ / 2

b. Effect of a red laser on the band spacing

- ☐ Spacing increases (longer wavelength, and $\Delta x = \lambda L/d$)

Answer / notes: The bands get further apart.

State average 0.7 / 1 (70%); 63% scored full marks. Mark distribution: 0: 37% · 1: 63%

Your mark: _____ / 1

c. How the experiment supports a wave model, not a particle model

- ☐ An interference pattern of alternating bright and dark bands is produced
- ☐ Interference (superposition, constructive/destructive by path difference) is a wave property
- ☐ A particle model predicts only two bright bands, one behind each slit

Answer / notes: Interference is a wave-only effect. Particles would give two bands.

State average 1.9 / 3 (63%); 36% scored full marks. Mark distribution: 0: 20% · 1: 11% · 2: 33% · 3: 36%

Your mark: _____ / 3

Question 7 (6 marks) · state average 3.5 / 6 (58%)

2023 VCAA exam · Section B · Question 14

a. Speed of the neutrons

☐ $v = \frac{h}{m\lambda}$ with substitution

☐ $v \approx 1.3 \times 10^3 \text{ m s}^{-1}$

Answer / notes: $\frac{6.63 \times 10^{-34}}{1.67 \times 10^{-27} \times 3.02 \times 10^{-10}} = 1.31 \times 10^3 \text{ m s}^{-1}$

State average 1.4 / 2 (70%); 67% scored full marks. Mark distribution: 0: 26% · 1: 7% · 2: 67%

Your mark: _____ / 2

b. Will a crystal of spacing $3.62 \times 10^{-10} \text{ m}$ diffract the beam?

☐ Yes

☐ The wavelength is similar to the spacing ($\lambda/w \approx 0.8$, close to 1), so noticeable diffraction occurs

Answer / notes: Yes. λ and spacing are of the same order.

State average 0.8 / 2 (40%); 18% scored full marks. Mark distribution: 0: 43% · 1: 39% · 2: 18%

Your mark: _____ / 2

c. Electron or neutron with the same wavelength: which is faster?

☐ The electron

☐ Same λ means same momentum $p = mv$; the electron's much smaller mass means a greater speed

Answer / notes: The electron.

State average 1.3 / 2 (65%); 58% scored full marks. Mark distribution: 0: 31% · 1: 12% · 2: 58%

Your mark: _____ / 2

Question 8 (8 marks) · state average 4.9 / 8 (61%)

2023 VCAA exam · Section B · Question 15

a. Work function of calcium from the graph

☐ 3 eV (magnitude of the y-intercept; positive)

Answer / notes: 3 eV

State average 0.7 / 1 (70%); 67% scored full marks. Mark distribution: 0: 33% · 1: 67%

Your mark: _____ / 1

b. Maximum wavelength that ejects electrons

☐ Threshold frequency from the x-intercept $\approx 7.2 \times 10^{14} \text{ Hz}$ (or $f_0 = \phi/h$)

☐ $\lambda_{max} = c/f_0 \approx 4.1 \times 10^{-7} \text{ m}$

Answer / notes: About $4.1 \times 10^{-7} \text{ m}$.

State average 1.0 / 2 (50%); 47% scored full marks. Mark distribution: 0: 50% · 1: 3% · 2: 47%

Your mark: _____ / 2

c. Draw the line for copper (work function 4.70 eV)

☐ Parallel to the calcium line (same gradient, h)

☐ Shifted right: intercepts -4.70 eV on the y-axis (threshold $\approx 11.4 \times 10^{14} \text{ Hz}$)

Answer / notes: A parallel line, further right.

State average 1.5 / 2 (75%); 69% scored full marks. Mark distribution: 0: 22% · 1: 9% · 2: 69%

Your mark: _____ / 2

d. Does 380 nm light eject electrons from copper?

☐ Photon energy $= hc/\lambda \approx 3.3 \text{ eV}$ (or $f \approx 7.9 \times 10^{14} \text{ Hz}$)

☐ Compared with the 4.70 eV work function (or the copper threshold frequency)

☐ No: the photon energy is less than the work function, so no photoelectrons

Answer / notes: No. 3.3 eV is less than 4.70 eV.

State average 1.7 / 3 (57%); 51% scored full marks. Mark distribution: 0: 40% · 1: 34% · 2: 5% · 3: 51%

Your mark: _____ / 3

Question 9 (6 marks) · state average 3.8 / 6 (63%)

2023 VCAA exam · Section B · Question 16

a. Photon energy for the 436.6 nm line

☐ $E = hc/\lambda$ with substitution

☐ $E \approx 2.84 \text{ eV}$

Answer / notes: $1.24 \times 10^{-6} / 436.6 \times 10^{-9} = 2.84 \text{ eV}$

State average 1.4 / 2 (70%); 70% scored full marks. Mark distribution: 0: 28% · 1: 2% · 2: 70%

Your mark: _____ / 2

b. Draw the transition for the 436.6 nm line

☐ Downward arrow from $n = 4$ (7.73 eV) to $n = 2$ (4.89 eV)

Answer / notes: 7.73 → 4.89 eV, pointing down.

State average 0.6 / 1 (60%); 58% scored full marks. Mark distribution: 0: 42% · 1: 58%

Your mark: _____ / 1

c. All photon energies emitted after absorbing 6.7 eV

☐ 6.70 eV ($n = 3 \rightarrow 1$)

☐ 4.89 eV ($n = 2 \rightarrow 1$)

☐ 1.81 eV ($n = 3 \rightarrow 2$)

Answer / notes: 6.70, 4.89 and 1.81 eV.

State average 1.8 / 3 (60%); 54% scored full marks. Mark distribution: 0: 35% · 1: 6% · 2: 4% · 3: 54%

Your mark: _____ / 3

Question 10 (7 marks) · state average 4.2 / 7 (60%)

2024 VCAA exam · Section B · Question 3

a. Height of the cliff (landing 6.2 s after being hit)

☐ Vertical: $u_y = 48 \sin 35^\circ = 27.5 \text{ m s}^{-1}$ upwards

☐ $s = u_y t - \frac{1}{2} g t^2$ with a consistent sign convention

☐ $h \approx 18 \text{ m}$

Answer / notes: $27.5(6.2) - 4.9(6.2)^2 = -17.7 \text{ m}$, so the cliff is about 18 m high.

State average 1.5 / 3 (50%); 37% scored full marks. Mark distribution: 0: 32% · 1: 23% · 2: 8% · 3: 37%

Your mark: _____ / 3

b. Horizontal distance travelled

☐ $x = 48 \cos 35^\circ \times 6.2$

☐ $x \approx 244 \text{ m}$

Answer / notes: 244 m

State average 1.3 / 2 (65%); 60% scored full marks. Mark distribution: 0: 38% · 1: 2% · 2: 60%

Your mark: _____ / 2

c. How air resistance would change the flight path

☐ Lower maximum height

☐ Shorter horizontal distance (range)

Answer / notes: Lower and shorter. Most students forgot the lower height.

State average 1.4 / 2 (70%); 48% scored full marks. Mark distribution: 0: 16% · 1: 36% · 2: 48%

Your mark: _____ / 2

Question 11 (4 marks) · state average 2.4 / 4 (60%)

2024 VCAA exam · Section B · Question 4

a. Impulse on the dummy in the initial test

☐ Impulse = area under the force–time graph (time converted to seconds)

☐ $\approx 2.9 \times 10^3 \text{ N s}$

Answer / notes: $\frac{1}{2} \times 24\,000 \times 0.240 \approx 2.9 \times 10^3 \text{ N s}$

State average 1.0 / 2 (50%); 47% scored full marks. Mark distribution: 0: 52% · 1: 2% · 2: 47%

Your mark: _____ / 2

b. Sketch the force–time graph with the airbag (double the duration)

- ☐ Duration doubled (480 ms)
- ☐ Same area (same impulse), so the peak force is halved ($\approx 12\,000\text{ N}$)

Answer / notes: Same area, twice as long, half as high.

State average 1.4 / 2 (70%); 51% scored full marks. Mark distribution: 0: 18% · 1: 31% · 2: 51%

Your mark: _____ / 2

Question 12 (3 marks) · state average 1.7 / 3 (57%)

2024 VCAA exam · Section B · Question 6

Evaluate the suggestion with calculations

- ☐ Gravitational force available: $F = \frac{Gm_1m_2}{r^2} \approx 1.8 \times 10^{-6}\text{ N}$
- ☐ Centripetal force required: $\frac{mv^2}{r} = \frac{1.50 \times 10^4 \times 0.436^2}{500} \approx 5.7\text{ N}$
- ☐ No: the gravity available is far smaller than the force required

Answer / notes: No. About 10^{-6} N is available but about 6 N is needed.

State average 1.7 / 3 (57%); 43% scored full marks. Mark distribution: 0: 28% · 1: 17% · 2: 13% · 3: 43%

Your mark: _____ / 3

Question 13 (6 marks) · state average 3.4 / 6 (57%)

2024 VCAA exam · Section B · Question 7

a. Show the electron's speed is $1.32 \times 10^7\text{ m s}^{-1}$

☐ $v = \frac{qBr}{m} = \frac{1.60 \times 10^{-19} \times 5.00 \times 10^{-3} \times 0.0150}{9.11 \times 10^{-31}} = 1.32 \times 10^7$

Answer / notes: Radius in metres (0.0150 m).

State average 0.8 / 1 (80%); 73% scored full marks. Mark distribution: 0: 27% · 1: 73%

Your mark: _____ / 1

b. Accelerating voltage needed

- ☐ $qV = \frac{1}{2}mv^2$
- ☐ $V \approx 4.96 \times 10^2\text{ V}$

Answer / notes: $\frac{9.11 \times 10^{-31} \times (1.32 \times 10^7)^2}{2 \times 1.60 \times 10^{-19}} = 496\text{ V}$

State average 1.2 / 2 (60%); 55% scored full marks. Mark distribution: 0: 38% · 1: 7% · 2: 55%

Your mark: _____ / 2

c. Path of a proton at the same speed, justified

- ☐ Curves the opposite way (positive charge, so the force is reversed)
- ☐ Much larger radius
- ☐ Because $r = mv/qB$ and the proton's mass is about 1800 times larger (same charge magnitude)

Answer / notes: Opposite direction, far bigger radius.

State average 1.4 / 3 (47%); 27% scored full marks. Mark distribution: 0: 23% · 1: 41% · 2: 10% · 3: 27%

Your mark: _____ / 3

Question 14 (8 marks) · state average 4.3 / 8 (54%)

2024 VCAA exam · Section B · Question 15

a. Momentum of an electron in the beam

- ☐ $p = h/\lambda$
- ☐ $p \approx 3.3 \times 10^{-24}\text{ kg m s}^{-1}$

Answer / notes: $6.63 \times 10^{-34} / 2.00 \times 10^{-10} = 3.3 \times 10^{-24}$

State average 1.4 / 2 (70%); 67% scored full marks. Mark distribution: 0: 29% · 1: 4% · 2: 67%

Your mark: _____ / 2

b. Who is right about a detectable diffraction pattern?

- ☐ Max
- ☐ The wavelength and interatomic spacing are of the same order of magnitude ($\lambda/w \approx 0.7$), so there will be useful diffraction

Answer / notes: Max. Similar order of magnitude is enough.

State average 0.7 / 2 (35%); 26% scored full marks. Mark distribution: 0: 60% · 1: 14% · 2: 26%

Your mark: _____ / 2

c. How electrons and X-rays can give almost identical patterns

- ☐ The pattern spacing depends on wavelength (λ/w)
- ☐ So the electrons and X-ray photons must have the same wavelength

Answer / notes: Same wavelength.

State average 0.9 / 2 (45%); 23% scored full marks. Mark distribution: 0: 33% · 1: 44% · 2: 23%

Your mark: _____ / 2

d. Energy of the X-ray photons in eV

- ☐ $E = hc/\lambda$ with $\lambda = 5.01 \times 10^{-11}$ m
- ☐ $E \approx 2.48 \times 10^4$ eV

Answer / notes: 24 800 eV (much more than the electrons' 600 eV).

State average 1.3 / 2 (65%); 61% scored full marks. Mark distribution: 0: 32% · 1: 7% · 2: 61%

Your mark: _____ / 2

Question 15 (6 marks) · state average 3.2 / 6 (53%)

2025 VCAA exam · Section B · Question 8

a. Show the period is 6.62×10^3 s

- ☐ $r = 6.37 \times 10^6 + 1.25 \times 10^6 = 7.62 \times 10^6$ m
- ☐ $T = 2\pi\sqrt{r^3/GM} = 6.62 \times 10^3$ s

Answer / notes: Add the altitude to Earth's radius, in metres.

State average 1.3 / 2 (65%); 50% scored full marks. Mark distribution: 0: 23% · 1: 27% · 2: 50%

Your mark: _____ / 2

b. Gravitational force on the satellite

- ☐ $F = \frac{GMm}{r^2}$ with $r = 7.62 \times 10^6$ m
- ☐ $F \approx 1.79 \times 10^4$ N

Answer / notes: 1.79×10^4 N

State average 0.9 / 2 (45%); 41% scored full marks. Mark distribution: 0: 51% · 1: 7% · 2: 41%

Your mark: _____ / 2

c. Why the satellite accelerates towards Earth (Newton's laws)

- ☐ The only (unbalanced) force on it is Earth's gravity, directed towards Earth
- ☐ By Newton's second law ($a = F/m$) it accelerates in the direction of the net force, changing the direction of its velocity (centripetal acceleration)

Answer / notes: Net force towards Earth, so acceleration towards Earth.

State average 1.0 / 2 (50%); 33% scored full marks. Mark distribution: 0: 30% · 1: 37% · 2: 33%

Your mark: _____ / 2

Question 16 (7 marks) · state average 4.3 / 7 (61%)

2025 VCAA exam · Section B · Question 12

a. Why the flux through the coil changes as it rotates

- ☐ The angle between the coil and the field changes, so the perpendicular component (area facing the field) changes

Answer / notes: Its orientation to the field changes.

State average 0.6 / 1 (60%); 65% scored full marks. Mark distribution: 0: 36% · 1: 65%

Your mark: _____ / 1

b. Show the flux change from horizontal to vertical is 7.5×10^{-4} Wb

☐ $\Delta\Phi = BA = 0.60 \times (0.050 \times 0.025) = 7.5 \times 10^{-4}$ Wb

Answer / notes: Uses the full area.

State average 0.7 / 1 (70%); 74% scored full marks. Mark distribution: 0: 26% · 1: 74%

Your mark: _____ / 1

c. Average EMF over the quarter turn at 50 Hz

☐ $\Delta t = T/4 = 0.005$ s

☐ $\varepsilon = N \frac{\Delta\Phi}{\Delta t}$

☐ $\varepsilon = 3.0$ V

Answer / notes: $20 \times 7.5 \times 10^{-4} / 0.005 = 3.0$ V. Using 0.02 s was the common error.

State average 1.9 / 3 (63%); 56% scored full marks. Mark distribution: 0: 23% · 1: 19% · 2: 2% · 3: 56%

Your mark: _____ / 3

d. Change to produce DC output, with reason

☐ Replace the slip rings with a split-ring commutator

☐ It reverses the coil's connections every half turn, so the output keeps the same polarity

Answer / notes: A split-ring commutator.

State average 1.1 / 2 (55%); 36% scored full marks. Mark distribution: 0: 29% · 1: 35% · 2: 36%

Your mark: _____ / 2

Question 17 (3 marks) · state average 1.7 / 3 (57%)

2025 VCAA exam · Section B · Question 16

a. Planck's constant from the graph

☐ Gradient of the trend line (rise/run, using points on the line)

☐ $h \approx 4.2 \times 10^{-15}$ eV s (values close to this accepted)

Answer / notes: Gradient of the line, about 4.2×10^{-15} eV s.

State average 1.1 / 2 (55%); 40% scored full marks. Mark distribution: 0: 32% · 1: 29% · 2: 40%

Your mark: _____ / 2

b. Work function from the graph

☐ ≈ 2.2 eV (magnitude of the extrapolated y-intercept)

Answer / notes: 2.2 eV

State average 0.6 / 1 (60%); 56% scored full marks. Mark distribution: 0: 44% · 1: 56%

Your mark: _____ / 1

Question 18 (3 marks) · state average 1.7 / 3 (57%)

2025 VCAA exam · Section B · Question 17

Rachel's speed relative to Sandra, as a multiple of c

☐ Proper time is Rachel's 1.000 s: $\gamma = 1.010/1.000 = 1.010$

☐ $v = c\sqrt{1 - 1/\gamma^2}$

☐ $v \approx 0.140c$

Answer / notes: $\gamma = 1.010$, so $v = 0.140c$. γ can never be less than 1.

State average 1.7 / 3 (57%); 51% scored full marks. Mark distribution: 0: 33% · 1: 12% · 2: 4% · 3: 51%

Your mark: _____ / 3

Your result

Total: _____ / 122 State average on these questions: **72 / 122 (59%)**

Topics to revise:

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